

Kaden Lee

EDUCATION

University of California, San Diego	09/2025 - 06/2026
MS in Mechatronics and Controls	GPA: 3.82/4.0
University of California, Santa Barbara	10/2020 - 06/2024
BS in Mechanical Engineering	GPA: 3.74/4.0

WORK EXPERIENCE

FoodTools	07/2024 - 07/2025
Jr. Mechanical Design Engineer	Santa Barbara, CA
- Designed custom parts and assemblies for automated food portioning machines in SolidWorks - Refined the machining process of intricate parts using Fusion 360 Manufacturing (CAM) - Drafted and updated engineering drawings in SolidWorks - Supported the assembly team in wiring and assembling electrical panels, pneumatic control systems, and machinery	

Introba	12/2023 - 06/2024
New Technologies Research Intern	Oakland, CA (Remote)
- Contributed to an interdisciplinary team of engineers to develop the UCSB Clean Energy Master Plan - Researched energy storage technologies and HVAC systems for a planned central utility plant - Developed calculators that modeled the energy needs of buildings around the UCSB campus in Microsoft Excel - Designed a seawater heat exchange system in SolidWorks	

UCSB Mechanical Engineering Machine Shop	12/2022 - 12/2023
Remedial Tutor	Santa Barbara, CA
- Supervised 8 students twice per week in the Mechanical Engineering Machine Shop - Taught students how to use manual lathes, CNC mills, drill presses, and measurement tools - Guided students in the fabrication of parts for a compressed air motor	

Meissner	06/2023 - 09/2023
Research and Development Intern	Camarillo, CA
- Conducted research on membrane porosity characterization - Developed a test method to determine porosity of membranes across varying porosities - Designed and machined laboratory equipment using manual mills and lathes - Performed dynamic mechanical analysis to determine the effects of solvents on mechanical properties of epoxies - Completed formal documentation, technical reports, and presentations to share results and data	

PROJECTS

Terrain Simulator	01/2026 - 03/2026
Graduate Student	San Diego, CA
- Designed and built a 5-bar kinesthetic lower-limb haptic device for simulating variable terrain underfoot - Integrated BLDC motors, VESC motor controllers, incremental encoders, and a Teensy 4.1 microcontroller into a complete electromechanical hardware system - Programmed 5 static and 2 dynamic virtual environments using impedance control in Arduino IDE - Analyzed joint torques in Python to achieve 15 lb of end effector resistance across the desired workspace - Identified and resolved Hall-effect sensor faults using a DMM and oscilloscope	

Senior Capstone Project	09/2023 - 06/2024
Undergraduate Student	Santa Barbara, CA
- Developed an assistive mobility device for young children with cerebral palsy or other severe motor impairments - Integrated BLDC motors, an Arduino Mega, and various sensors into the electromechanical system - Designed, modeled, and drafted engineering drawings with GD&T for multiple components using SolidWorks - Utilized finite element analysis (FEA) to evaluate mechanical components of the system under various loads - Fabricated major mechanical components using techniques including milling, water jetting, and 3D printing	

CERTIFICATIONS AND COURSES

Coding for Everyone: C and C++ Specialization (UCSC / Coursera)	03/2025 - 05/2025
Machine Learning with Python (IBM / Coursera)	03/2025 - 04/2025

TECHNICAL SKILLS

Skills: MATLAB, SolidWorks, CAM, Igor, COMSOL, SolidWorks Simulation, Python, C/C++, Microsoft Office